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BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT



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(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to
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DEPARTMENT OF CSE (DATA SCIENCE)

A Mini-Project Report On

“Sales Report Analysis Dashboard Using Power BI”

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

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Visvesvaraya Technological University

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CERTIFICATE

This is to certify that the OPEN-ENDED PROJECT of DATA VISUALIZATION WITH POWER BI “**Sales Report Analysis Dashboard Using Power BI**” has been successfully presented by **VINUTHA V** bearing USN **3BR23CD105** student of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide

Mr. Sundeep Sindhanoor

Signature of HOD

Dr. Aradhana D

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1. Introduction

1.1 About the Project Sales analysis is a critical component of modern business intelligence, as it helps organizations understand performance, identify trends, and make informed decisions. In many companies, large volumes of sales data are generated daily, but without proper tools, this data remains underutilized. This project focuses on developing an **interactive Sales Report Analysis Dashboard using Power BI**, which transforms raw data into meaningful insights through visualization techniques. The dashboard provides a complete overview of business performance by analyzing sales, budget comparisons, customer contributions, product performance, and geographic distribution.

The project consists of three main dashboards:

- **Sales Overview Dashboard** – Provides a summary of overall performance
- **Customer Details Dashboard** – Focuses on customer-level analysis
- **Product Details Dashboard** – Highlights product performance.

1.2 Language Used The implementation of this project involves several tools and technologies that work together to process and visualize data effectively.

- **Power BI Desktop:** Used for creating dashboards and visual reports. It provides drag-and-drop features to build interactive visuals.
- **Power Query:** Used for data cleaning and transformation. It helps in removing duplicates, handling missing values, and formatting the dataset.
- **DAX (Data Analysis Expressions):** Used to create calculated measures such as total sales, budget comparison, and rankings.
- **Dataset (Sales Data):** Contains structured data including customer details, product information, sales values, and locations.

2 Scope of the Project

2.1 Problem Statement In many organizations, sales data is stored in raw formats such as spreadsheets, making it difficult to analyze effectively. Traditional reports are often static and do not provide interactive insights. As a result, decision-makers face challenges in identifying trends, comparing sales performance with targets, and understanding customer or product behaviour. Organizations generate large volumes of sales data, but analyzing this data effectively using traditional methods is difficult. Static reports do not provide

clear insights into performance, trends, or comparisons with targets. This makes it challenging for decision-makers to identify key business areas such as top customers, best-selling products, and regional performance. Therefore, there is a need for an interactive and visual system that can convert raw data into meaningful insights. This project addresses the problem by developing a Power BI dashboard for efficient sales analysis. This project addresses these challenges by creating an interactive Power BI dashboard that converts raw sales data into meaningful and visually appealing insights, enabling better analysis and decision-making.

2.2 Objectives:

- **Data Preparation and Cleaning:** To import the sales dataset into Power BI and clean it using Power Query. This ensures the data is accurate, consistent, and ready for analysis.
- **Sales Performance Analysis:** To calculate key metrics such as total sales, budget, and the difference between them. This helps in evaluating whether business targets are being achieved.
- **Customer and Product Analysis:** To identify top-performing customers and products based on sales data. This helps businesses focus on high-value customers and profitable products.
- **Interactive Dashboard Development:** To create an interactive dashboard with filters and slicers. This allows users to explore data dynamically and gain better insights for decision-making.

3 System Requirements

3.1 Functional Requirements The functional requirements define what the system must be able to do in order to achieve the project objectives.

- Importing data from Excel or CSV files into Power BI
- Cleaning and transforming data using Power Query
- Creating DAX measures for calculations such as total sales and differences
- Generating visualizations including charts, maps, and tables
- Providing filters and slicers for user interaction
- Displaying a complete dashboard for analysis

3.2 Non-Functional Requirements Non-functional requirements ensure the system is efficient, reliable, and easy to use.

- **Usability:** The dashboard should be simple and easy to understand
- **Performance:** Visuals should load quickly and respond efficiently
- **Scalability:** The system should support additional data in the future
- **Accuracy:** All calculations must be precise and reliable.

3.3 Hardware Requirements To run Power BI effectively and handle CSV datasets, the following hardware is recommended:

- A computer or laptop with at least **8 GB RAM**.
- Processor: Intel i5 or equivalent (minimum dual-core).
- Storage: At least 500 GB HDD/SSD for dataset storage and software installation.
- Display: Full HD resolution (1920x1080) for clear visualization of dashboards.

3.4 Software Requirements The software environment required for the project includes:

- **Power BI Desktop:** Used to create interactive sales dashboards and analyze business performance.
- **Power Query:** Used to clean and transform sales data before visualization.
- **DAX:** Used to calculate sales metrics such as total sales, budget, and performance indicators.
- **Sales Dataset (Excel/CSV):** Used as the input data containing customer, product, and sales details for analysis.

4 Methodology

4.1 Data Collection The dataset used in this project consists of sales-related information including customer names, product details, sales amount, budget values, and customer locations. The data also includes time-based fields such as year and month, which are used to analyze trends. This structured dataset enables detailed analysis of sales performance across different dimensions like customers, products, and regions.

4.2 Data Preparation (Power Query) The dataset was imported into Power BI and cleaned using Power Query. Missing values were handled, duplicate records were removed, and column names were standardized for better understanding. Data types such as numerical values for sales and date formats for time analysis were correctly assigned. These steps ensured that the dataset was accurate and ready for visualization.

4.3 Data Modelling and Calculations

After cleaning, the data was organized into a structured model. DAX measures were created to calculate key metrics such as:

- Total Sales
- Total Budget
- Sales Difference (Sales vs Budget)

These calculations form the base of KPI cards shown in your dashboard (e.g., overall sales difference value displayed at the top).

4.4 Data Visualization (Based on Your Dashboards)

Sales Overview Dashboard - The main dashboard provides a summary of business performance:

- KPI card displays Sales vs Budget difference (e.g., 45,695)
- Donut chart shows sales distribution by product category (Furniture, Technology, Office Supplies)
- Bar charts highlight Top 5 Customers and Top 5 Products
- Line chart shows sales trends across months (2019–2021)
- Map visual displays sales distribution across different cities

This dashboard gives a quick overview of overall sales performance.

Customer Details Dashboard - This dashboard focuses on customer-level analysis:

- KPI cards show total sales and budget values
- Line chart shows monthly sales trends for customers
- Table (heatmap format) displays customer-wise sales distribution across months
- Bar chart identifies Top 10 customers based on sales

- Map shows customer locations and their contribution to sales

This helps in identifying high-value customers and understanding their behaviour.

Product Details Dashboard - This dashboard analyzes product performance:

- KPI cards compare sales vs budget for products
- Bar chart shows Top 10 products by sales
- Additional charts display product category and subcategory performance
- Map visual represents product sales across regions

This helps in identifying best-selling and low-performing products.

4.5 Filters and Interactivity - To make the dashboard interactive, slicers were added based on:

- Year (2019, 2020, 2021)
- Month
- Customer City
- Product Category and Subcategory
- Product Name

These filters allow users to explore specific data segments and perform customized analysis, making the dashboard dynamic.

4.6 Dashboard Design

The dashboard is designed with a clear and logical layout. KPI cards are placed at the top to provide quick insights into performance. Charts such as bar graphs, line charts, and donut charts are arranged below to show detailed analysis. Map visuals are used to represent geographic data. A consistent color scheme is applied across all dashboards for better readability and visual appeal.

4.7 Outcome

The final outcome is a fully interactive Power BI dashboard consisting of Sales Overview, Customer Details, and Product Details pages. It provides insights into sales trends, customer contributions, product performance, and regional distribution. The dashboard helps users easily identify performance gaps, top contributors, and areas for improvement, supporting effective decision-making.

5 Implementation

5.1 Dataset Description

The dataset used in this project contains detailed sales information including customer names, product categories, subcategories, product names, sales amount, budget values, and customer locations. It also includes time-related fields such as year and month, which are used to analyze sales trends over a period of time. This dataset enables comprehensive analysis of sales performance across different dimensions like customer, product, and region.

5.2 Data Loading

The dataset was imported into Power BI Desktop using Excel/CSV format. Power BI provides a simple interface for loading data, and after importing, the dataset was checked to ensure all columns and values were correctly loaded. This step ensures that the data is ready for cleaning and further processing.

5.3 Data Transformation (Power Query)

Data transformation was performed using Power Query to prepare the dataset for analysis. Duplicate records were removed to maintain accuracy, and missing values were handled appropriately. Unnecessary columns were removed to simplify the dataset. Column names were renamed for better readability, and correct data types such as numeric, text, and date formats were assigned. These steps ensured that the data was clean, consistent, and suitable for building the dashboard.

5.4 Data Modelling

After cleaning, the dataset was structured into a proper format within Power BI. Fields such as customer, product, and sales were logically arranged to support analysis. Even though the dataset may be in a single table, it was organized to ensure smooth interaction between visuals and accurate calculations across the dashboard.

5.5 DAX Measures (Key Calculations)

DAX (Data Analysis Expressions) was used to create important measures required for analysis. Measures such as Total Sales and Total Budget were calculated to represent actual and target values. A Sales vs Budget Difference measure was created to identify performance gaps. These measures are displayed in KPI cards and used across various charts to provide meaningful

insights into business performance.

5.6 Dashboard Visualization

The dashboard was created using different types of visuals to represent data effectively:

- KPI Cards show overall sales, budget, and their difference
- Bar Charts display top customers and top products based on sales
- Line Charts represent monthly sales trends over multiple years
- Donut Charts show distribution of sales across product categories
- Map Visuals display geographic sales distribution across cities

These visuals help in understanding data quickly and clearly.

5.7 Filters / Slicers

Interactive slicers were added to enhance the usability of the dashboard. Users can filter the data based on year, month, customer city, product category, subcategory, and product name. These slicers allow users to focus on specific data and perform detailed analysis, making the dashboard dynamic.

5.8 Dashboard Layout

The dashboard is designed in a structured way with three main sections: Sales Overview, Customer Details, and Product Details. KPI cards are placed at the top for quick insights, while charts and maps are arranged below for detailed analysis. This layout improves readability and helps users easily navigate through the dashboard.

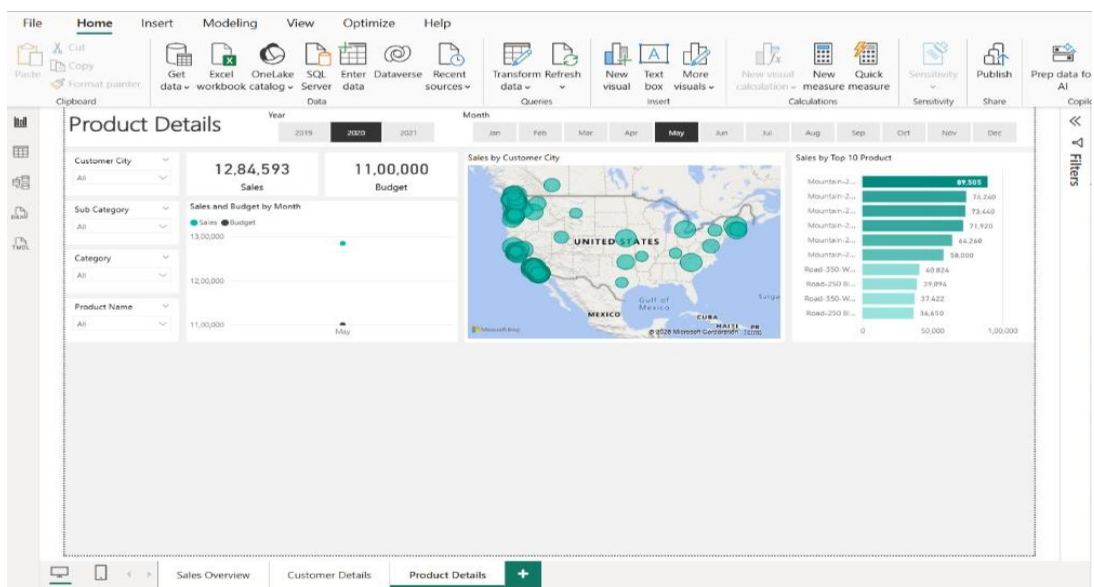
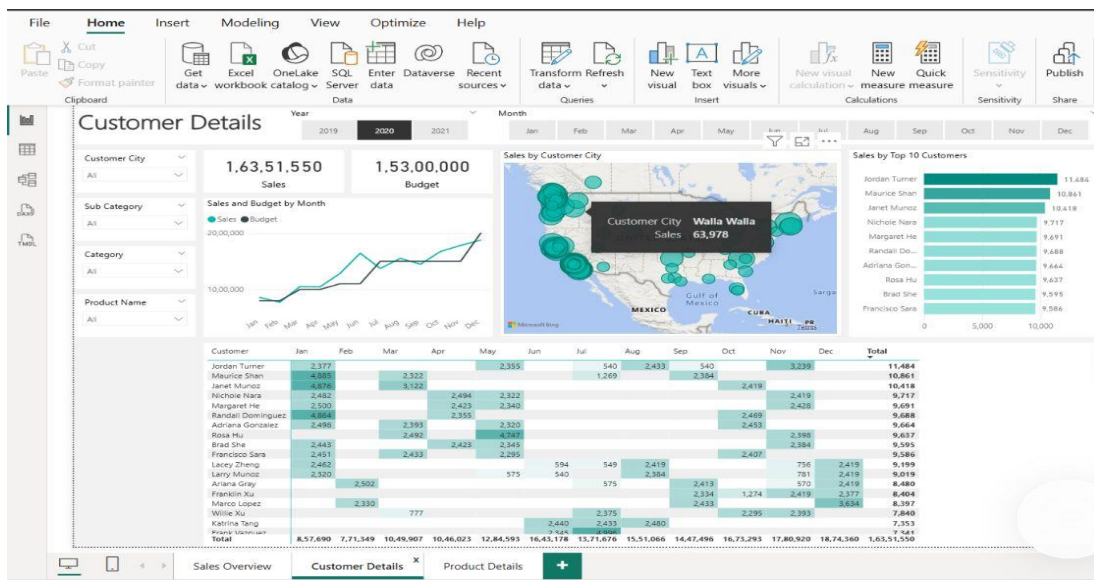
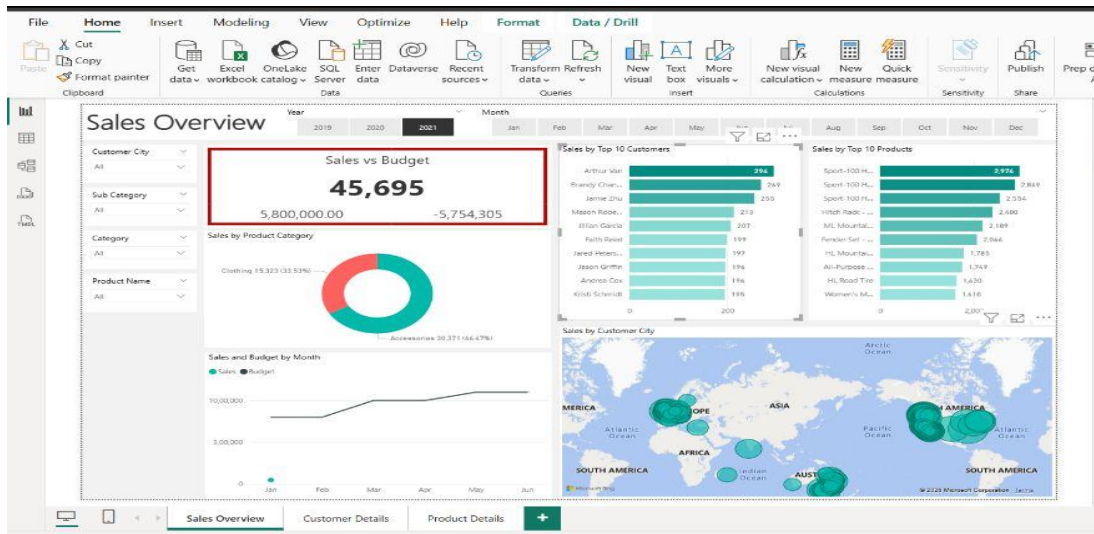
5.9 Integration

All visuals in the dashboard are interconnected using Power BI's cross-filtering feature. When a user selects a specific value in one visual, all other visuals update automatically. This integration allows users to drill down into the data and analyze it from different perspectives without switching between pages.

5.10 Final Output

The final output is a fully interactive Power BI dashboard that provides a complete view of sales performance. It shows sales vs budget comparison, identifies top customers and products, and highlights geographic trends. The dashboard helps users make better decisions by providing clear and meaningful insights.

Result



Conclusion

This project successfully demonstrates the effective use of Power BI in transforming raw sales data into meaningful and actionable insights. By systematically following the steps of data collection, data cleaning using Power Query, and applying DAX measures, the project was able to analyze key business metrics such as total sales, budget values, and performance differences. These calculations provided a clear understanding of how well the business is performing against its targets.

The development of three interactive dashboards—**Sales Overview, Customer Details, and Product Details**—played a crucial role in presenting the analysis in a structured and user-friendly manner. The Sales Overview dashboard provided a high-level summary of performance, while the Customer Details dashboard helped in identifying top customers and understanding their contribution to revenue. Similarly, the Product Details dashboard enabled analysis of product performance, highlighting best-selling and low-performing items. The inclusion of maps further enhanced the analysis by showing geographic distribution of sales.

One of the key strengths of this project is the use of interactivity through slicers and filters, which allows users to explore data dynamically based on year, month, customer, and product categories. This feature makes the dashboard more flexible and useful for real-time analysis. The visual elements such as KPI cards, bar charts, line charts, and donut charts simplify complex data and make it easier for users to understand patterns and trends.

In conclusion, the Sales Report Analysis Dashboard serves as a powerful tool for analyzing business performance and can be further extended in the future by integrating larger datasets, real-time data sources, or advanced analytics techniques.

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